

Teacher Answer Key

Mechanical Oscillations – Ex 6: Oscillator on Inclined Plane

1. Calculation of Stiffness k

At equilibrium position O, the sum of forces acting on (S) is zero:

$$\vec{W} + \vec{R} + \vec{T} = \vec{0}.$$

Projecting on the x-axis (parallel to the slope, downwards):

$$W_x - T = 0$$

$$mg \sin(\alpha) - k\Delta l_0 = 0$$

With $\Delta l_0 = l_{eq} - l_0 = 25 - 20 = 5 \text{ cm} = 0.05 \text{ m}$.

$$k = \frac{mg \sin(\alpha)}{\Delta l_0}$$

$$k = \frac{0.1 \times 10 \times \sin(30^\circ)}{0.05} = \frac{1 \times 0.5}{0.05} = \frac{0.5}{0.05}$$

Result: $k = 10 \text{ N/m}$

2. Mechanical Energy

a) Expression of Mechanical Energy (ME):

Reference of GPE : Horizontal plane through O ($x = 0$).

For a point at abscissa x (down the slope), the vertical height is $h = -x \sin(\alpha)$.

1. $GPE = mgh = -mgx \sin(\alpha)$

2. $EPE = \frac{1}{2}k(\text{deformation})^2$. The deformation is $\Delta l = \Delta l_0 + x$.

$$EPE = \frac{1}{2}k(\Delta l_0 + x)^2 = \frac{1}{2}k\Delta l_0^2 + k\Delta l_0x + \frac{1}{2}kx^2$$

$$ME = KE + GPE + EPE$$

$$ME = \frac{1}{2}mv^2 - mgx \sin(\alpha) + \frac{1}{2}k\Delta l_0^2 + k\Delta l_0x + \frac{1}{2}kx^2$$

Since $mg \sin(\alpha) = k\Delta l_0$ (from equilibrium), the terms $-mgx \sin(\alpha)$ and $+k\Delta l_0x$ cancel out.

Result: $ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \text{Constant}$

(Constant = $\frac{1}{2}k\Delta l_0^2$)

b) Velocity Verification:

Conservation of Energy between state (x, v) and max displacement $(X_m, 0)$:

$$ME = \frac{1}{2}kX_m^2 + C = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + C$$

$$\frac{1}{2}mv^2 = \frac{1}{2}k(X_m^2 - x^2) \implies v^2 = \frac{k}{m}(X_m^2 - x^2)$$

We know $\frac{k}{m} = \frac{g \sin(\alpha)}{\Delta l_0}$.

$$v = \pm \sqrt{\frac{g \sin(\alpha)}{\Delta l_0}(X_m^2 - x^2)}$$

3. Kinematics

a) Calculate V_0 :

At $t = 0$, $x = x_0 = 5 \text{ cm} = 0.05 \text{ m}$ and $X_m = 8 \text{ cm} = 0.08 \text{ m}$.

$$V_0^2 = \frac{k}{m}(X_m^2 - x_0^2) = \frac{10}{0.1}(0.08^2 - 0.05^2)$$

$$V_0^2 = 100(0.0064 - 0.0025) = 100(0.0039) = 0.39$$

$$V_0 = \sqrt{0.39} \approx 0.624 \text{ m/s}$$

Result:

$$V_0 \approx 0.62 \text{ m/s}$$

b) Max Speed Position & Value:

The speed is maximum when the oscillator passes through the equilibrium position ($x = 0$).

$$V_{max} = \omega X_m = \sqrt{\frac{k}{m}} X_m$$

$$V_{max} = \sqrt{100} \times 0.08 = 10 \times 0.08$$

Result:

$$V_{max} = 0.8 \text{ m/s}$$

c) Differential Equation & Period:

Derivative of ME : $mvx'' + kxv = 0 \implies x'' + \frac{k}{m}x = 0$.

$$\text{Period } T_0 = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.1}{10}} = 2\pi\sqrt{0.01} = 0.2\pi.$$

Result:

$$T_0 \approx 0.63 \text{ s}$$

4. Equation of Motion

Form: $x = a \sin(bt + c)$. This corresponds to $x = X_m \sin(\omega t + \varphi)$.

a) Finding a, b, and c:

1. $a = X_m = 8 \text{ cm} = 0.08 \text{ m}$

2. $b = \omega = 10 \text{ rad/s}$

3. Finding c (φ):

At $t = 0$, $x = 5 \text{ cm}$.

$$5 = 8 \sin(c) \implies \sin(c) = \frac{5}{8} = 0.625$$

Two possible mathematical values for c in $[0, 2\pi)$:

- $c_1 \approx 0.68 \text{ rad}$ (Quadrant 1)
- $c_2 = \pi - 0.68 \approx 2.46 \text{ rad}$ (Quadrant 2)

Selection: The block is launched with speed $\vec{V} = V_0 \vec{i}$ (downward, positive x).
So $v_0 > 0$.

$$v = x' = a \cdot b \cos(bt + c)$$

At $t = 0$, $v_0 = a \cdot b \cos(c)$. Since $a, b, v_0 > 0$, we need $\cos(c) > 0$.

Only c_1 (Quadrant 1) has a positive cosine.

Result: $c \approx 0.68 \text{ rad}$

b) Graph:

Sine wave starting at $x = 5 \text{ cm}$, going up to 8 cm , then oscillating.

